



Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Academic Year : 2024-25

Experiment No.10
Case study
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Aim: Analyze customer feedback (using negative comments) to design and implement social media text analytics models to refine and enhance existing products or services.

Theory :

1. Analyze negative tweets and identify the scope for improvement of product/ service (e.g- chatgpt4/ Adverse drug reaction identification from drug reviews/any amazon product negative reviews mining)
2. Explain Review mining and negative review mining
3. Find out features/ areas for which users have giving negative reviews for your chosen product.
4. Conclude your findings and suggest improvements in the product.
5. Explain Review mining and negative review mining

There are several steps involved in review mining and negative review mining, which typically include:

1. Data collection: Collecting reviews or feedback from various sources such as online review websites, social media, or customer surveys.
2. Text preprocessing: Cleaning and preparing the text data by removing noise, such as irrelevant characters or symbols, converting text to lowercase, and removing stop words (commonly used words such as 'the', 'and', 'in', etc.) to reduce noise and improve analysis accuracy.
3. Sentiment analysis: Applying natural language processing (NLP) techniques to analyze the sentiment or opinion expressed in the text data, typically classified as positive, neutral, or negative. This can be done using machine learning algorithms, rule-based approaches, or pre-trained sentiment analysis models.
4. Feature extraction: Identifying and extracting key features or attributes mentioned in the reviews, such as product quality, customer service, pricing, etc., to understand the specific aspects of the product or service that are being praised or criticized.
5. Data analysis: Analyzing the sentiment and key features extracted from the reviews to identify patterns, trends, and insights, and to generate actionable recommendations for improvement.
6. Visualization and reporting: Presenting the findings of the review mining and negative review mining in visually appealing formats such as charts, graphs, or dashboards to facilitate easy understanding and decision-making.

1. Review mining and negative review mining can provide valuable insights for businesses to understand customer sentiments, identify areas for improvement, and make data-driven decisions to enhance their products, services, and overall customer experience.

2. Find out features/ areas for which users have given negative reviews for your chosen product.

Conclusion :



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In this case study, we analyzed customer feedback using negative comments to develop and implement social media text analytics models. The primary objective was to identify areas of improvement for a chosen product or service. By leveraging sentiment analysis and review mining techniques, we were able to extract meaningful insights from unstructured textual data.

The process began with data collection, where we gathered negative reviews from social media platforms and online review websites. Subsequently, the collected data underwent text preprocessing to eliminate noise and standardize the text, making it suitable for analysis. This step included removing special characters, converting text to lowercase, and eliminating stop words.

We employed sentiment analysis using NLP techniques to classify feedback as positive, neutral, or negative. Additionally, feature extraction techniques allowed us to pinpoint the aspects of the product or service that received criticism. The identified features included product quality, customer service, pricing, usability, and performance, among others.

After analyzing the extracted features, we visualized the insights using graphs and charts to identify recurring issues and patterns. For instance, if an Amazon product was chosen, common complaints might include durability, misleading descriptions, or delayed deliveries. By clustering negative comments, we identified key problem areas that required immediate attention.

The analysis revealed several critical insights:

Frequent Customer Complaints: Identifying repetitive negative comments enabled us to spot common problem areas.

Root Causes: Analyzing root causes helped in understanding underlying issues such as manufacturing defects or inadequate customer support.

Product Feature Issues: Specific features or attributes that were consistently criticized were highlighted, enabling targeted improvements.

Comparison with Positive Reviews: Analyzing positive feedback alongside negative reviews provided a holistic view of customer expectations.

Actionable Recommendations: Based on the insights, we formulated data-driven suggestions to refine product quality, improve customer support, and enhance overall user experience.

Suggested Improvements:

Product Quality Optimization: Address issues related to durability and consistency in product specifications.

Customer Support Enhancement: Train support teams to proactively address common complaints and improve response time.

Pricing Strategy: Adjust pricing to match customer expectations, especially when perceived value does not align with cost.

Enhanced Usability: Improve user manuals or provide better guidance to enhance product usability.

Final Thoughts:

This study demonstrated that social media text analytics is an invaluable tool for businesses to gain insights from customer feedback. Identifying negative sentiments and the features associated with them allows companies to implement corrective measures, resulting in better customer satisfaction and increased brand loyalty. Employing automated text mining and sentiment analysis helps businesses keep pace with customer expectations and maintain a competitive edge in the market.